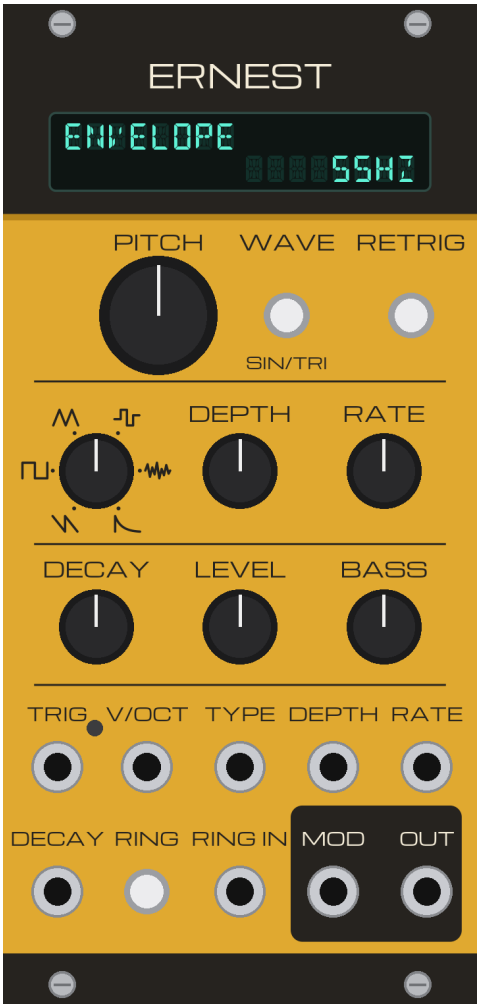
ERNEST

Percussion oscillator with six ways to bend its pitch

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Ernest is a single percussion voice built around an oscillator whose pitch can be bent six different ways. Hit it with a trigger and you get kicks, toms, snares, zaps and metallic pings. Leave the trigger unpatched and it plays continuously, which is handy for tuning.

Beyond the classic drum-machine voice it has a few modular extras: a pitch input, CV over the modulation shape, depth and rate and over the decay, a modulation output and a free-running mode.

AT A GLANCE

- Sine or triangle oscillator, 20 Hz to 12 kHz, with a 1 V/oct input
- Six pitch modulation shapes, from a falling saw to a pitch envelope
- Modulation from 0.1 Hz to 5 kHz: slow wobbles turn into new tone colours
- One-knob decay envelope, bass boost with drive, ring modulation
- 12 HP

CONTROLS

Screen	The modulation shape being played (knob plus TYPE input) and the base pitch.
PITCH	Base pitch, from 20 Hz to 12 kHz. It starts at 55 Hz. With V/OCT patched, it transposes the incoming note in semitones instead.
WAVE	Lit: triangle, slightly brighter. Off: sine.
RETRIG	Lit (default): every hit restarts the modulation, so every hit sounds the same. Off: the modulation keeps running and each hit lands somewhere else in its cycle.
Shape selector	The six icons around this knob: **saw down** (the pitch falls, again and again), **square** (two pitches alternate), **triangle** (rises and falls), **random** (a new random step each cycle), **noise** (bursts of noise in the pitch, for snares) and **envelope** (one sweep down after each hit, for kicks and toms).
DEPTH	How far the pitch moves, from −100 % to +100 %. Centre: no modulation. Negative values flip the direction. Gentle near the centre, up to ±8 octaves at the ends.
RATE	Modulation speed, from 0.1 Hz to 5 kHz. With the envelope shape, how fast the sweep falls back. Above about 20 Hz the modulation is heard as a new tone colour rather than movement.
DECAY	Length of the volume envelope, from 3 ms to 3 s.
LEVEL	Output level.
BASS	Boosts the low end, then drives the sound into saturation as you turn it up.
RING	Lit: the oscillator is multiplied by the signal at RING IN. Nothing comes out unless both are sounding.

INPUTS

TRIG	Plays a hit: restarts the oscillator, both envelopes and, with RETRIG lit, the modulation. The small light flashes on each hit. Unpatched, Ernest plays continuously.
V/OCT	Patched, Ernest plays the note it receives, 1 volt per octave with 0 V = C4 as everywhere in Rack, and the display shows the note. PITCH then transposes it in whole semitones from its starting position, so the notes stay in tune.
TYPE	Added to the shape selector, 1 V per shape: +1 V moves one icon clockwise, −1 V one icon back. Feed it a sequencer CV to change the sound from hit to hit.
DEPTH	Added to the DEPTH knob; ±5 V covers the whole range.
RATE	Added to the RATE knob; 1 V moves it a tenth of its travel.
DECAY	Added to the DECAY knob; 1 V moves it a tenth of its travel, roughly doubling or halving the length.
RING IN	The signal used by ring modulation (±5 V).

OUTPUTS

MOD	The pitch modulation itself, after DEPTH: 5 V for 8 octaves. Use it to move other modules in step with each hit.
OUT	Audio, ±5 V at full LEVEL.

HOW IT WORKS

Which way the pitch moves

Saw down, square, triangle and envelope start from PITCH and move up from it (or down, with negative DEPTH), then come back. Random and noise move both ways around PITCH.

The envelope jumps to full on every hit and falls back at a speed set by RATE. Noise is strongest at the start of each modulation cycle and fades towards its end, which gives snares their crack.

PATCH IDEAS

1. Analog kick

Ernest's starting settings: envelope shape, DEPTH 50 %, RATE at the centre, PITCH 55 Hz, DECAY around two o'clock. Patch a clock into TRIG.

2. Snare

Noise shape, WAVE lit, PITCH around 200 Hz, DEPTH 40 %, RATE far right, DECAY short, a little BASS.

3. Tuned toms

Envelope shape, DEPTH 30 %, a short DECAY, and a pitch sequence into V/OCT.

4. Metallic ping

Two Ernests triggered together. Patch the first OUT into the second RING IN and light RING on the second. Raise RATE on either for inharmonic overtones.

5. Zap

Envelope shape, DEPTH at 100 %, RATE low for a slow sweep, DECAY short.

SPECIFICATIONS

Width	12 HP
Pitch	20 Hz to 12 kHz, plus 1 V/oct input
Modulation	0.1 Hz to 5 kHz, up to ±8 octaves
Decay	3 ms to 3 s
Levels	Audio ±5 V. MOD output 5 V per 8 octaves. Trigger threshold about 1 V.